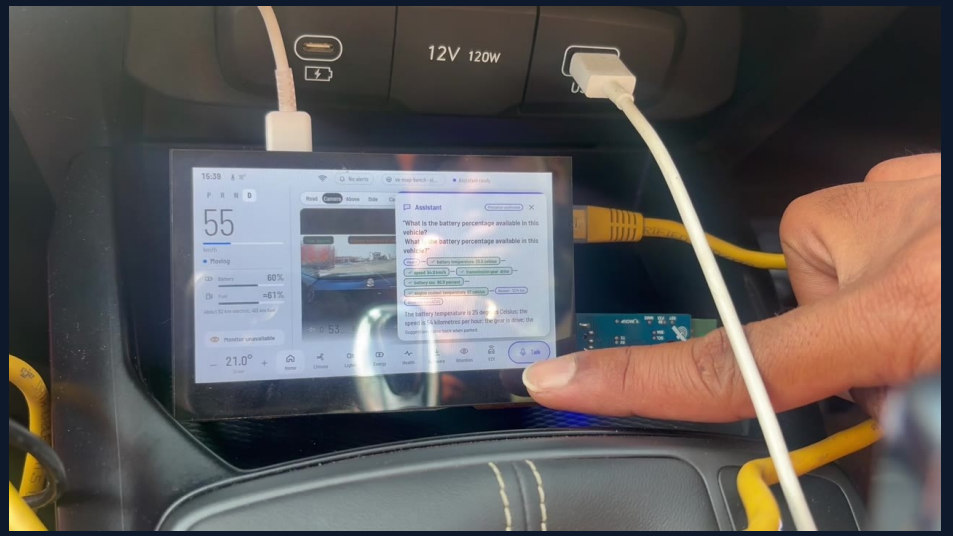
Ve-AP: a governed AI agent in a real car

Bounded authority and measured refusal for in-vehicle agents

Capstone project 2026 · Yuvaraja



No cloud · Jetson + Raspberry Pi 5 · tested in a car, 19 Oct 2026

Agents are arriving in cars. Three questions stay open

What may it do, and who decides?

A tool list says what an agent can call, not what it is allowed to call here, now, with this evidence.

How do we know a refusal works?

A safety rule that has never refused anything has never been shown to work.

How does it reach hardware without protocols?

If a model can emit a CAN ID, every guard above it can be bypassed by one string.

0

times a failure ceiling fired in 19 runs of an earlier evaluation

Not because it was rarely needed: the tests never created the event. A predicate with eight conditions has eight ways to repeat that mistake.

One predicate guards every vehicle call

$$\text{Allow}(q) = \text{Auth} \wedge \text{Schema} \wedge \text{Fresh} \wedge \text{Pre} \wedge \text{Risk} \leq \text{Permit} \wedge \text{Consent} \wedge \text{Budget} \wedge \text{Before}$$

Auth

in this agent's manifest?

`NOT_IN_MANIFEST`

Schema

arguments well formed?

`SCHEMA_INVALID`

Fresh

evidence young enough?

`STALE_EVIDENCE`

Pre

preconditions hold now?

`PRECONDITION_UNMET`

Risk ≤ Permit

class within the permit?

`RISK_EXCEEDS_PERMIT`

Consent

data use consented?

`REDACTION_UNAVAILABLE`

Budget

calls and cost left?

`BUDGET_EXHAUSTED`

Before

inside the deadline?

`DEADLINE_EXCEEDED`

Each conjunct must deny on its own, name itself, and leave evidence. All 8 are made to deny, with their own code, on every run.

L4 is the only way down

L9 Cloud / depot — optional, off

L8 Experience — cabin display, voice, console

L7 Orchestration — fast route, planner, budgets

L6 Specialised agent — read-only vehicle agent

L5 Shared intelligence — knowledge, driver state

L4 Capability gateway — Allow(q) · MCP · journal

L3 Vehicle services — SOVD north face over MHS

L2 Signals and transport — KUKSA · CAN

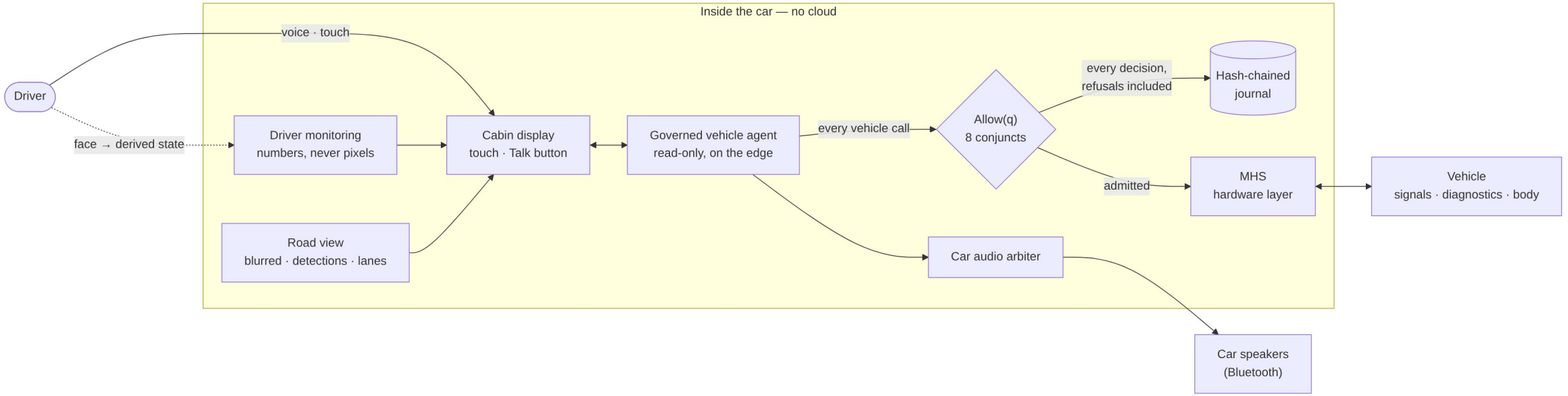
L1 Runtime — Jetson + Pi, containers, models

L0 Physical — ECUs, sensors, deterministic controllers

Invariants — each has a test that runs every time

- **I1** No model or agent touches a vehicle interface
- **I2** Only L3 adapters speak a protocol
- **I3** No CAN, service or UDS IDs in any capability
- **I4** High-rate data never crosses L4
- **I5** Deterministic controllers stay authoritative
- **I6** Only admitted paths actuate; only providers publish
- **I7** Isolation is a missing import, not a filter

The system on one page

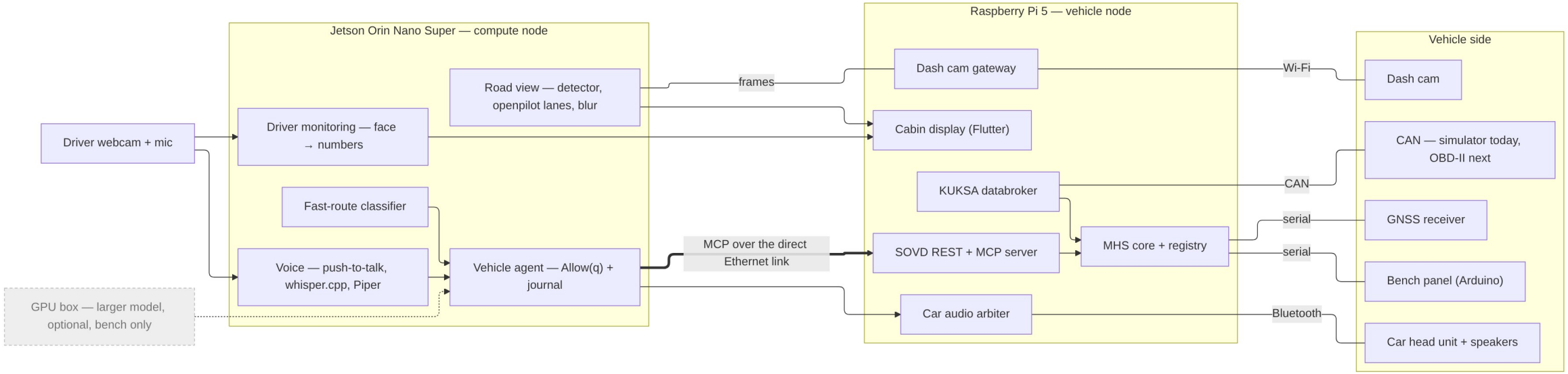


Every vehicle call goes through Allow(q)

Every decision, refusals included, is journalled

No cloud: two boards in the car

Two boards in the car, no cloud



Jetson Orin Nano Super

Vehicle agent, Allow(q) and journal · fast-route classifier · voice · driver monitoring · road view

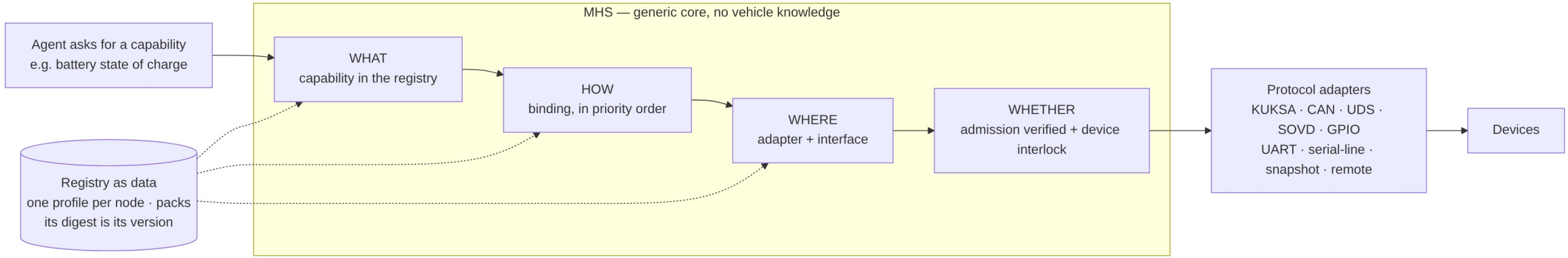
Raspberry Pi 5

MHS · SOVD + MCP server · KUKSA · CAN · cabin display · car audio · dash cam gateway

In the car

Dash cam · driver webcam + mic · GNSS · bench panel · the car's own speakers over Bluetooth

MHS: every hardware fact is checked data



Registry as data

Interfaces, capabilities, bindings, interlocks, lenses, controllers. Generated, checked, never hand-edited.

Pinned by digest

Nodes import each other pinned by digest. A mismatch answers REGISTRY_CHANGED, not a guess.

Honest fallback

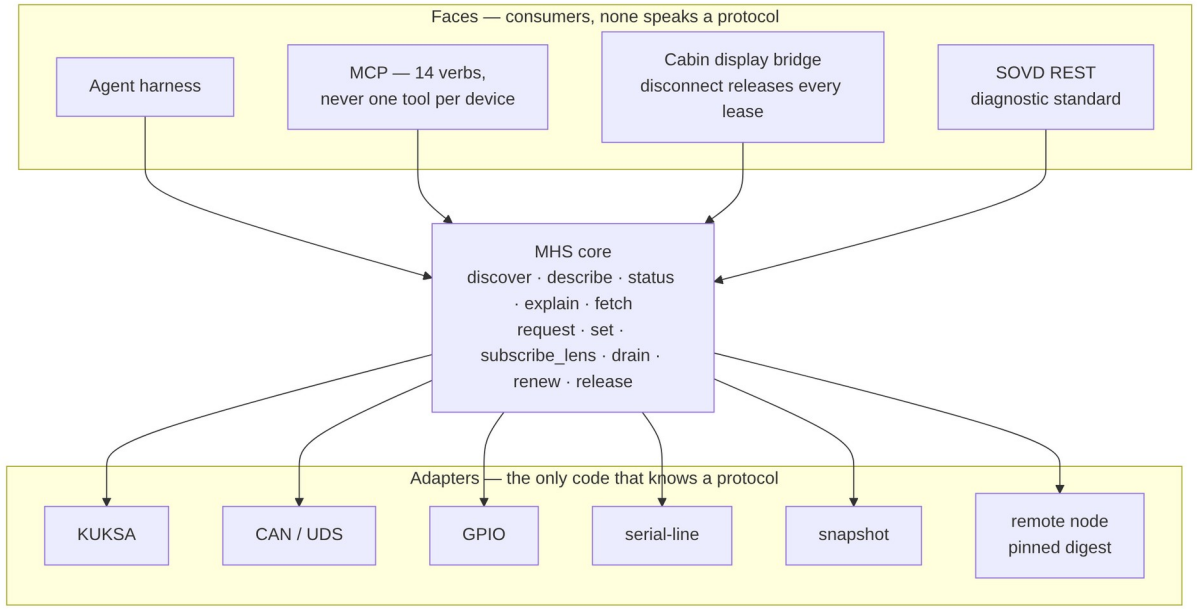
A read falls back only to the same quantity, as fresh or fresher. Writes never fall back.

Generic core

The core knows no vehicle, board or pin. A test fails if one appears in its logic.

MHS decides no authority: two locks, one core

Gate mints an admission for this exact request → **MHS** verifies it, and cannot mint one → **Interlock** the device policy may still refuse → **Adapter** the only code that speaks the protocol → **Read-back** the write is verified, then journalled



- Agent harness, MCP, cabin display and the SOVD diagnostic API are faces of one core. None speaks a protocol.
- Live data leaves only through budgeted lenses. Anything held is a lease that expires to a safe state.
- Faults are named: REGISTRY_CHANGED, INTERLOCK_REJECTED, LEASE_EXPIRED, CONTROLLER_HALTED.
- **481 MHS tests passed (19 Oct 2026).**

Alerts never wait for a model

Path A · deterministic, no model

Declared rules: coolant hot, door open while moving, belt unfastened

→ **Alert on screen and sound**

Path B · the governed agent

Through Allow(q), like every other call

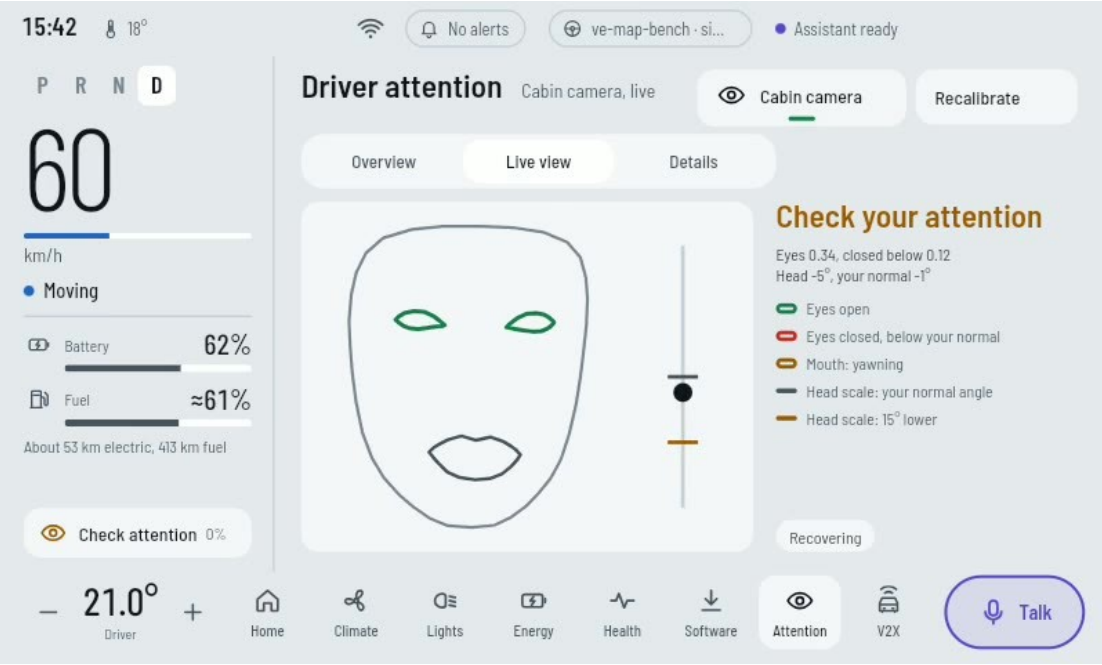
→ **May explain an alert. Never clear or suppress one**

Refused by name (C3)

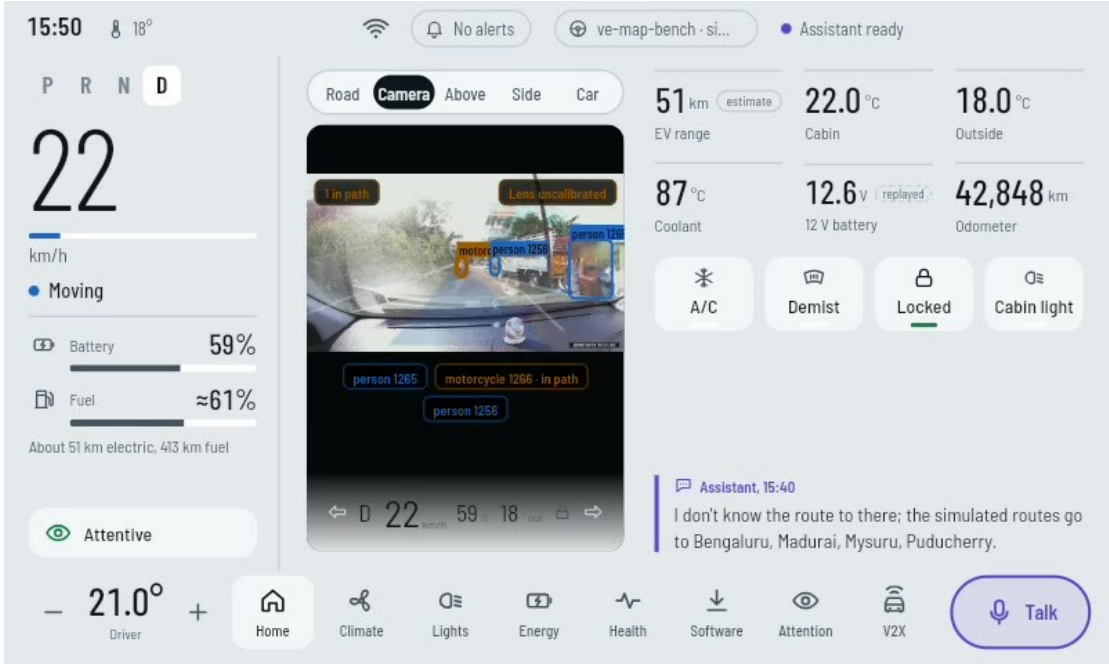
- `send_can_frame`
- `write_ecu_memory`
- `install_ota_bundle`
- `stream_forward_camera_raw`
- `get_cabin_frame`
- `identify_driver`
- `identify_speaker`
- `read_emotion`
- `detect_stress`
- `detect_smoking`
- `store_audio`

In no manifest, ever. The names exist only so a request for them is refused with NOT_IN_MANIFEST, before any model runs.

Privacy by construction, not by redaction



The driver's face, drawn from numbers



The dash cam, blurred before it leaves the computer

Driver camera
numbers out, never a frame

Dash cam
blurred, or no picture at all

Microphone
only while Talk is held

Location
only with consent

What is new

Allow(q), executed

8 conjuncts, each made to deny on every run

8/8

MHS hardware layer

hardware as checked data; cannot mint authority

481 tests

Refusal by name

safety-critical names exist only to be refused

built

Derived-state sensing

the driver camera emits numbers, never frames

in the car

Privacy-first road view

no blur means no picture

in the car

Two paths

alerts need no model; the agent never clears one

built

Model as last resort

~130 ms classifier (warm); 19.7 s → 0.86 s

12/12

Car audio arbiter

a critical alert pre-empts an answer; repeats are rate-limited

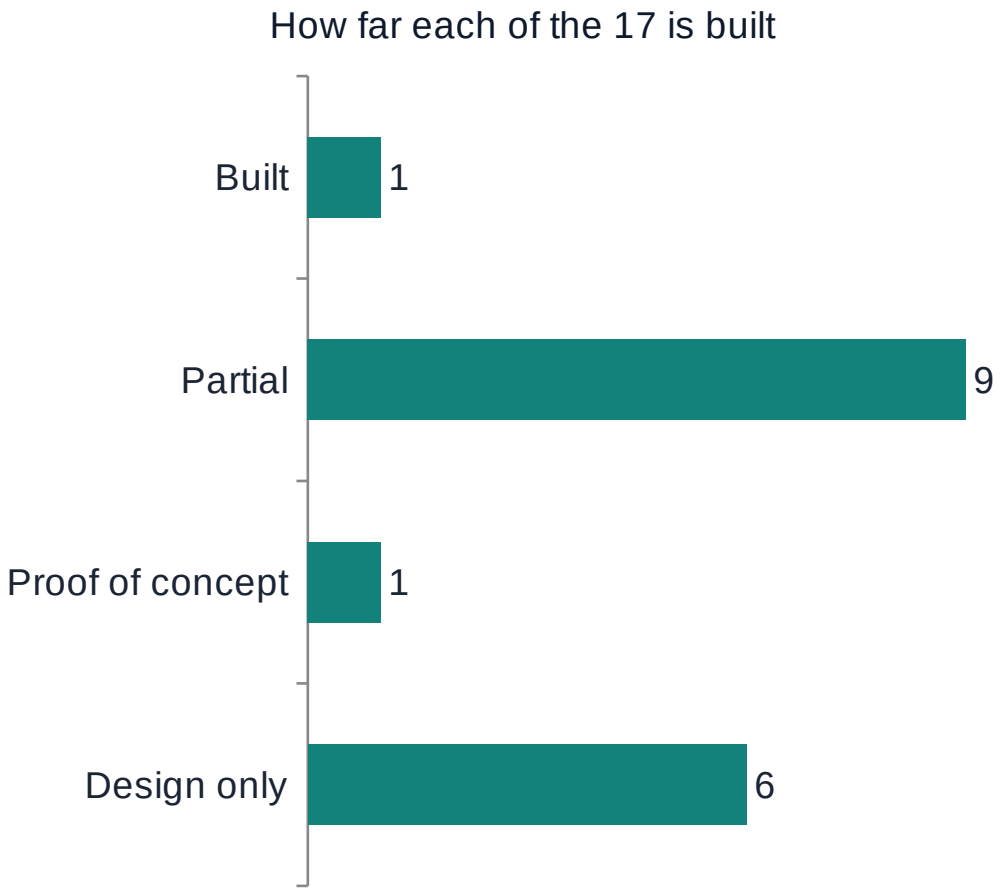
shown in car

Gated AI-SDLC

hooks stop the coding agent breaking an invariant

212 commits

Seventeen inventions, each with its real status



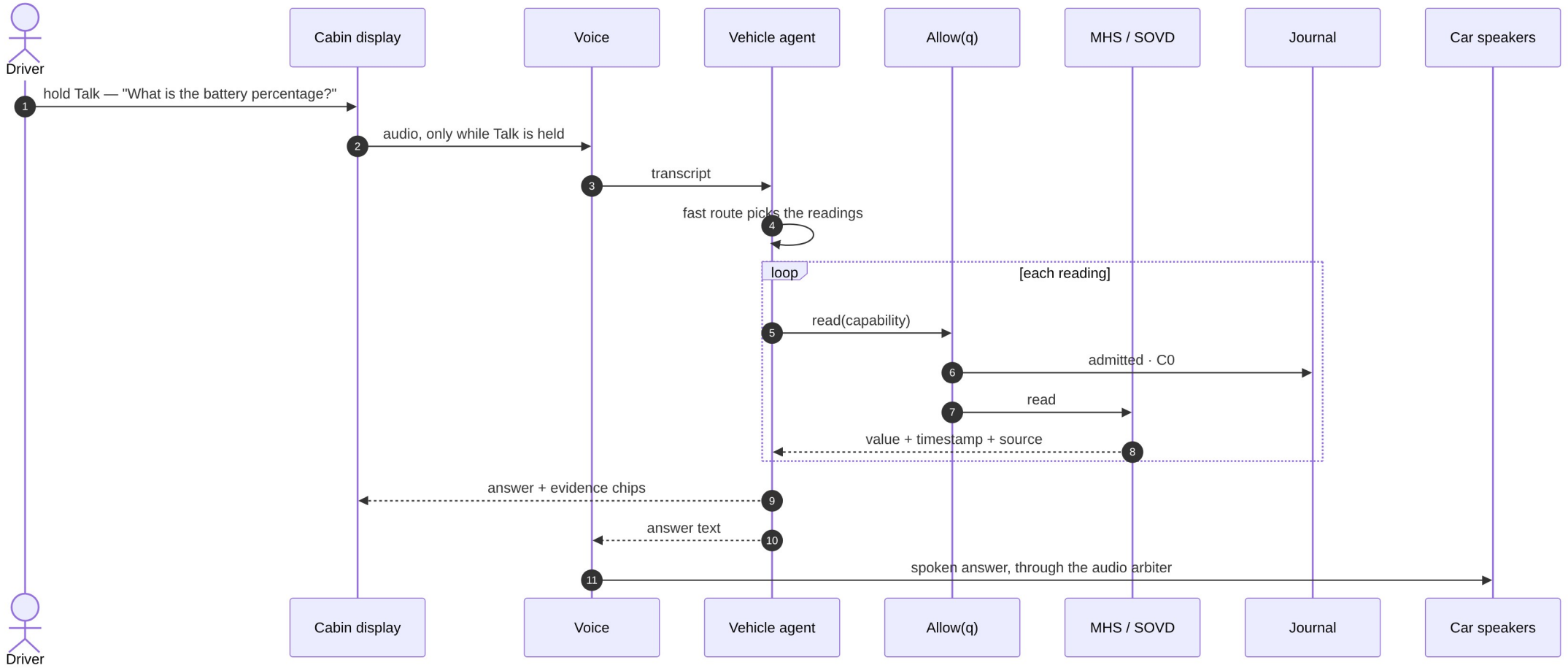
Built — 14 derived-state-only sensing

Partial — 1 authority-routed inference · 4 offline manifest · 6 evidence receipt · 8 A2UI in the car · 11 motion-state legality · 12 a generated tool is a proposal · 13 the plan is not the authority · 15 grammar-bounded calls · 17 presence = push-to-talk

Proof of concept — 2 the Speak predicate

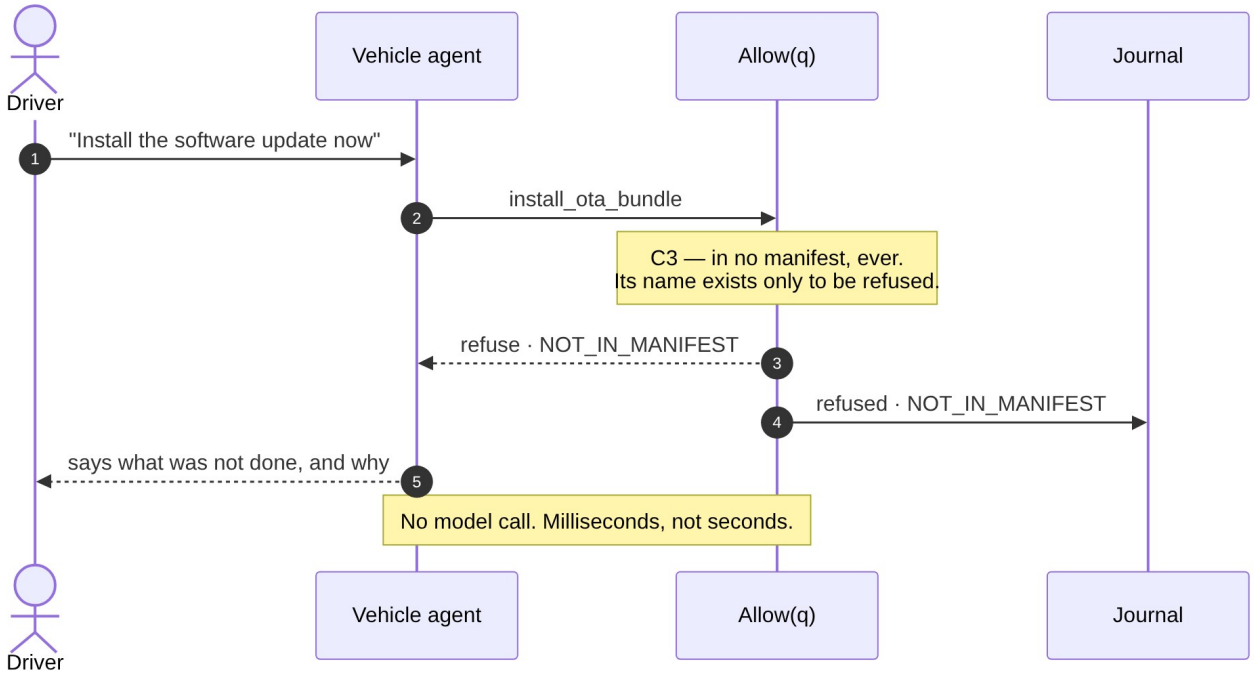
Design only — 3 refusal counterfactual · 5 joules per answer · 7 negative-answer accountability · 9 shadow verdict gate · 10 conflict arbitration · 16 signed model slots

Ask the car: every reading passes the gate



The fast route picks the readings. Each passes Allow(q) and is journalled. The answer shows its evidence and is spoken through the car's speakers.

Refusals are fast, named and journalled



Stale data is refused too

- A reading older than its bound is refused STALE_EVIDENCE.
- The agent says: "I could not read the coolant temperature — the reading is too old."
- Commands are refused before any model runs: milliseconds, not seconds.
- An old number is never shown as current.

The model is the last resort

1 **No model**

commands and C3 refused; template answers

2 **Small classifier, ~130 ms warm**

picks the one reading a question needs

3 **Larger model**

plans in a strict schema; every step re-enters the gate

If the larger model is unreachable: a model-free answer, or an honest "I can't answer that now".

19.7 → 0.86 s

median answer time, before and after the fast route

12 / 12

eval questions answered with the workstation model off (19 Oct)

0.4 s

median live answer, the same day

A test that has never failed proves nothing

Requirement	Test names its reason code	**Must fail for the stated reason**	Minimal code	Pass	Reachability	Mutation gate	Commit

Tiers — a tier that cannot run skips with its reason

- **T1** pure logic, any machine
- **T2** compute host
- **T3** CAN + ECU simulator
- **T4** governed: guard + journal
- **T5** two or more nodes
- **T6** shadow, recorded corpus

Reachability

Each conjunct is made to deny alone, with its own code and no other. 8/8.

Mutation gates

Break a rule on purpose: cabin, registry, audio, dev gates. A test must go red.

Traceability

Intent → requirement → plan → test → reason code. A gap breaks the build.

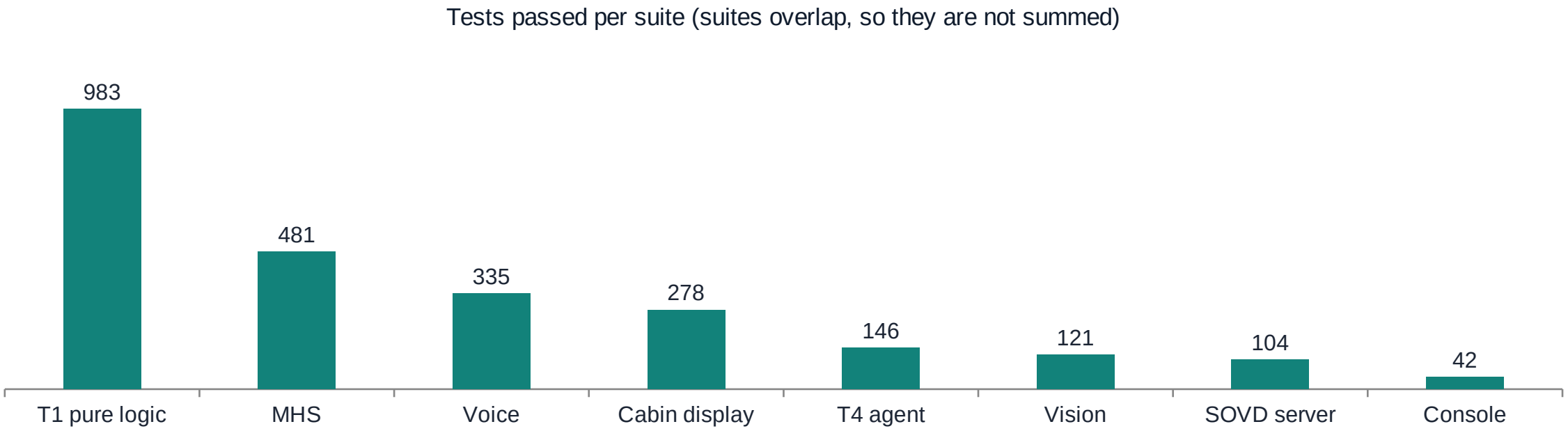
Results: 19 October 2026

18 / 18
check-all sections pass, 0 fail

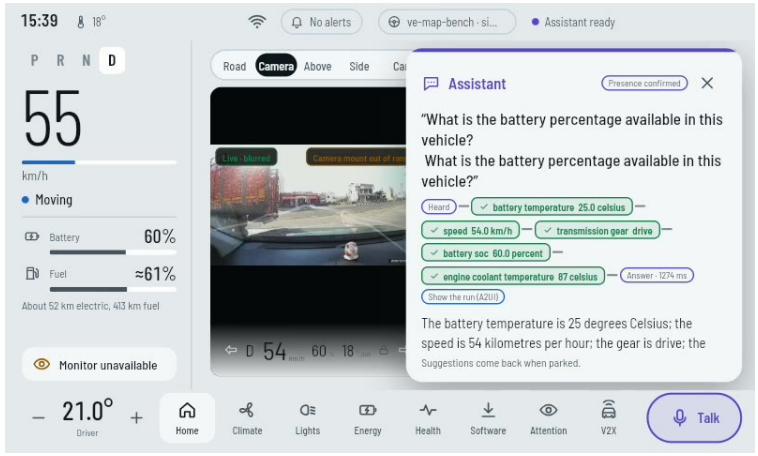
8 / 8
conjuncts deny with their own code

1,950
journal decisions, chain intact

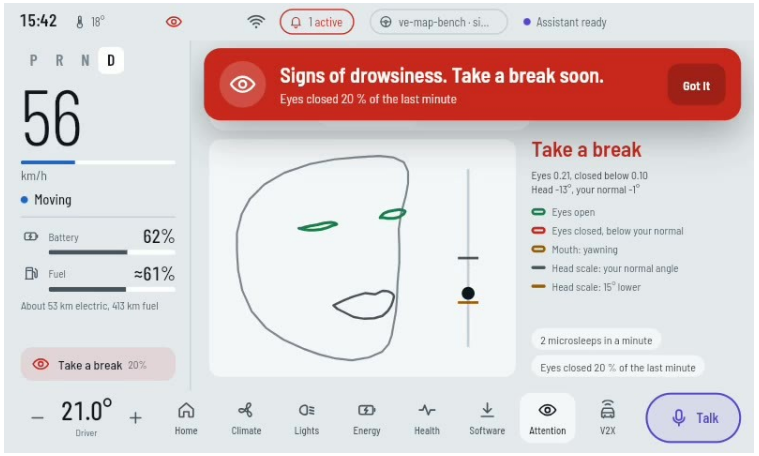
I1–I7
every invariant held



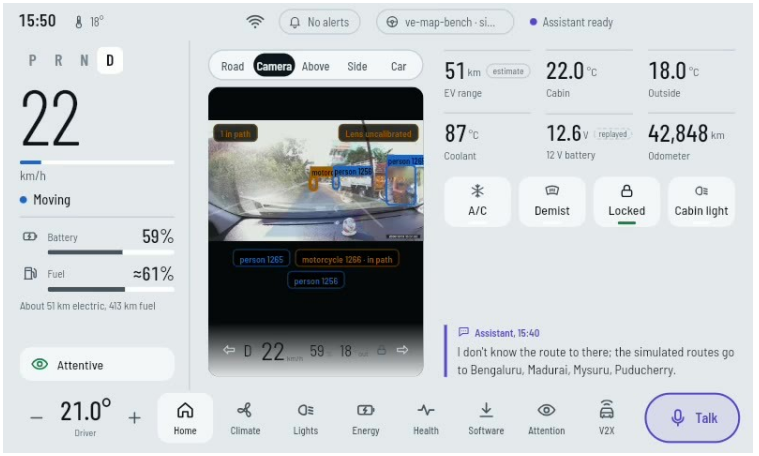
In the car: what was real, what was simulated



Spoken question, answer with its evidence



"Signs of drowsiness. Take a break soon."



Blurred dash cam, detections, in path

Real

Dash cam and road view · driver webcam and monitoring · microphone, speech, spoken answers · Bluetooth to the car's speakers · both boards on their own battery

Simulated

The vehicle signals the agent read: speed, battery %, coolant, fault codes, from a CAN simulator. The screen marks them estimate / replayed. The car's OBD-II port is the next stage.

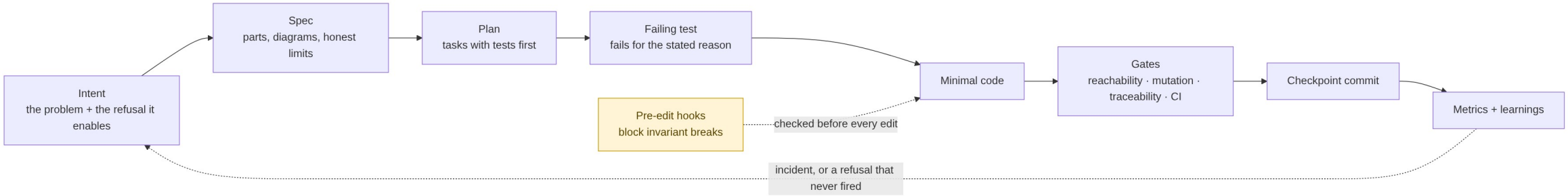
What the drive measured

	Drive 1	Drive 2	Drive 3 · camera straightened
Duration	5 min 41 s	29 min 5 s	6 min 53 s
Pictures processed	3,253	17,386	4,116
Rate (blur on)	9.8 fps	9.98 fps	9.97 fps
Camera mount	found at 42 s, lost at 64 s	out of range: "straighten the camera"	found after 74 s
Lane probability, left near	0.01	0.14	0.35
Lead car seen	0 %	1 %	13 %
Boxes flagged in path	0	601	149

It refused to calibrate a crooked camera and said why. Once the camera was straightened, the mount was found in 74 s.

Not done on the day: OBD-II in the car, lens calibration, a GNSS fix. Each is said, not hidden.

Built with an AI coding agent, inside gates



212

commits, 204 co-authored with the AI agent

54

intents · 47 specs · 10 plans

424

requirements traced to a test and a code (19 Oct)

7 weeks

3 Sep → 19 Oct 2026

Pre-edit hooks stop the coding agent before it writes:

a gate importing CAN or MCP · a camera frame escaping · a C3 capability · a flash path · an unsigned model slot · a push to main · a cloud SDK, or actuation outside an admitted path · protocol IDs in a capability. Each gate is probed both ways in CI.

Honest limits, and what comes next

Limits

- A simulated vehicle bus; OBD-II not yet in a car
- Live permit is C0 (read-only); C2 proven in a proof of concept
- Journal tamper-evident, not anchored or signed
- Driver monitoring not validated on drowsy drivers
- One car, one session, one operator
- No ISO 26262 or ISO/SAE 21434 claim

Next

1. OBD-II in the car: listen-only, then reads
2. Calibrate the lens; drive a marked road
3. C2 on the live path with a real writ issuer
4. Anchor the journal; sign the receipts
5. Wire the verdict-sequence shadow gate
6. Release MHS on its own, with its paper

Overclaiming is worth less than not claiming. Every limit above is also written in the documents.

Thank you

Documents, diagrams, slides and film: the ve-ap-capstone repository on GitHub

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